

Computer Architecture Lab

(EL-2012)

LABORATORY REPORT

Fall 2026



(Lab#03)

Apply Behavioral-level Verilog to Design and Simulate Combinational Modules

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Date performed: September 1, 2026

MARKS AWARDED: _____ / 10

Signature and date of Instructor: _____

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Updated: Fall 2026

Task 1: Write Verilog code for a 4-bit adder/subtractor. The circuit should perform addition or subtraction depending on the control signal. Simulate the results using a testbench.

Description:

In this task, we wrote the verilog code for a 4 bit adder/ subtractor. When the control line is 0, it performs addition and when the control line is 1, it performs subtraction. The results were further proven through the stimulation.

Code:

```
module bit_4_adder_subtractor(A,B,Control,result);
    input [3:0] A;
    input [3:0] B;
    input Control;
    output reg [3:0] result;
always @*
begin
    if (Control == 0)
        result = A + B;
    else
        result = A - B;
end
endmodule
module testbench();
reg [3:0] A;
reg [3:0] B;
reg Control;
wire [3:0] result;

bit_4_adder_subtractor uut(A, B, Control, result);

initial
begin

    A = 4'b0001;
    B = 4'b0011;
    Control = 0;
    #50
    A = 4'b0111;
    B = 4'b0110;
    #50 Control = 1;
end
endmodule
```

Output:

Wave - Default									
	Msgs								
sim:/testbench/A	0001	0001					0111		
sim:/testbench/B	0011	0011					0110		
sim:/testbench/Con...	0								
sim:/testbench/result	0100	0100					1101		

Wave - Default									
	Msgs								
sim:/testbench/A	0111	0001					0111		
sim:/testbench/B	0110	0011					0110		
sim:/testbench/Con...	1								
sim:/testbench/result	0001	0100					1101		

Task 2: Write Verilog code for 2x1 MUX having below mentioned operations. Simulate your program through testbench.

Note: Select, A and B are 1-bit inputs. Y is one bit output.

Select	Output (Y)
0	A&B
1	A B

Description:

In this lab, we wrote a verilog code using behavioral modeling for a 2x1 multiplexer. The circuit performs an and operation when (s==0) and or operation when (s==1) depending on the select signal. The results were further proven through the stimulation.

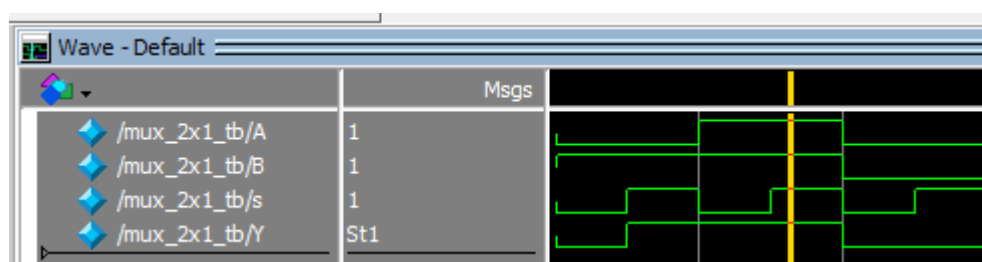
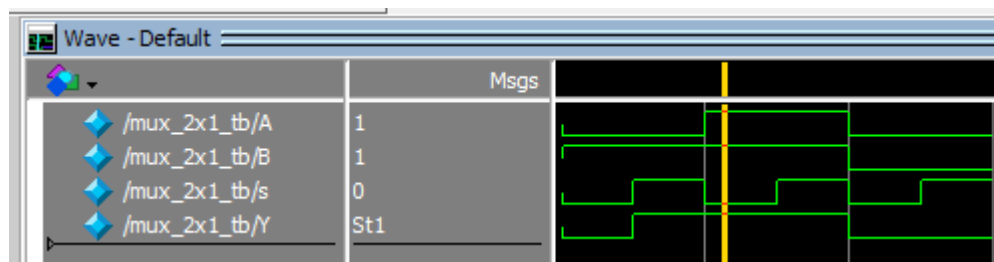
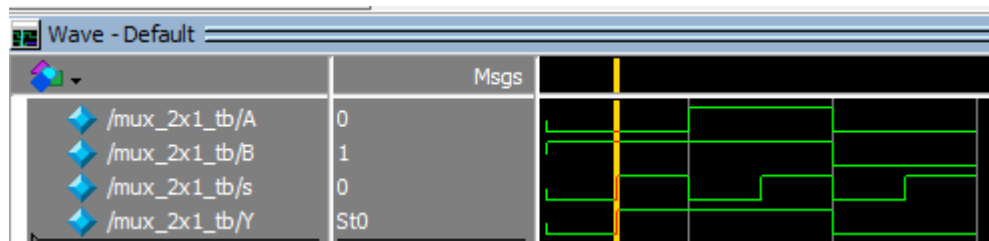
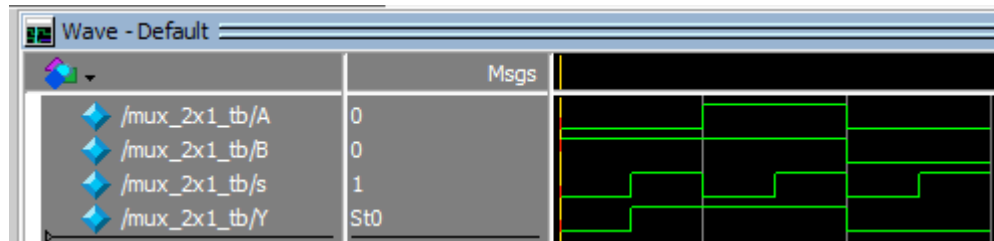
Code:

```
module mux_2x1(A,B,Y,s);
input A;
input B;
input s;
output reg Y;

always @*
begin
if(s==0)
Y=A&B;
else
Y=A|B;
end
endmodule

module mux_2x1_tb;
reg A,B,s;
wire Y;
mux_2x1 uut(A,B,Y,s);
initial
begin
A=0;B=1;s=0;
#50 A=0;B=1;s=1;
#50 A=1;B=1;s=0;
#50 A=1;B=1;s=1;
#50 A=0;B=0;s=0;
#50 A=0;B=0;s=1;
#50;
end
endmodule
```

Output:



Task 3: Write Verilog code for a 2-bit comparator that compares two 2-bit binary numbers and outputs the result as equality, greater than, or less than. Simulate the circuit with a testbench.

Description:

In this lab, we wrote a verilog code using behavioral modeling for a 2-bit comparator. The circuit compares two 2 bit inputs and determines whether they are equal, greater, or smaller. It was implemented and further verified through simulation.

Code:

```
module comparator_2bit (A, B, X, Y, Z);
    input [1:0] A;
    input [1:0] B;
    output reg X;
    output reg Y;
    output reg Z;

    always @*
    begin
        X = 1'b0;
        Y = 1'b0;
        Z = 1'b0;
        if (A == B)
            X = 1'b1;
        else if (A > B)
            Y = 1'b1;
        else
            Z = 1'b1;
    end
endmodule

module tb_comparator_2bit;
    reg [1:0] A;
    reg [1:0] B;
    wire X;
    wire Y;
    wire Z;
    comparator_2bit uut (A, B, X, Y, Z);
    initial
    begin
        A = 2'b00; B = 2'b01;
        #50; A = 2'b01; B = 2'b01;
        #50; A = 2'b11; B = 2'b01;
        #50;
    end
endmodule
```

Output:

